Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
B		**2511372**

Sustaining Juneau's Tomorrow -A System Dynamics Insight

Summary

In the context of the booming tourism industry in Juneau, Alaska, this paper develops a **Sustainable Tourism System Dynamics Model**, addressing the challenges of balancing tourism growth, environmental sustainability, and residents' well-being. This work provides a comprehensive analysis, including model construction, sensitivity analysis, and policy optimization, to achieve sustainable development goals.

We established a **Sustainable Tourism System Dynamics Model** based on the **RESET framework** (Resources, Environment, Society, Economics, Tourism). The model integrates key variables such as **Tourist Volume (TV)**, **Tourism Revenue (TR)**, **Resident Happiness Index (RHI)**, and **Environmental Degradation Index (EDI)**. Using Vensim software, causal relationships and system flow charts were constructed, and functions for critical variables were developed through regression analysis and theoretical assumptions. The model successfully simulates the dynamic interactions between tourism, environment, and society in Juneau.

A comprehensive **sensitivity analysis** was conducted to identify the most influential parameters, such as tax rates, road capacity, and environmental governance investment. Optimization strategies were proposed, including raising the tax rate to 0.17, increasing road capacity to 6,500 vehicles per hour, and allocating 20% of tourism revenue to environmental governance. These adjustments significantly improved **residents' well-being** (RHI increased by 15%) while reducing environmental degradation and maintaining tourism revenue growth.

We explored the model's adaptability to other overtourism regions and less-visited destinations. For overtourism areas, we proposed targeted strategies such as flow control and ecological thresholds, while for less-visited regions, measures like differentiated promotions and infrastructure investments were recommended. The model was successfully adapted to mountainous tourist areas by introducing variables like **Geological Disaster Risk Index** and **Carbon Emissions Index**, demonstrating its flexibility and applicability.

Policy recommendations were provided for the Juneau Tourism Board, including **dynamic tax adjustments**, **off-season incentives**, and **real-time monitoring systems** to ensure sustainable tourism development. Simulation results showed that these measures effectively balance economic growth, environmental protection, and social well-being.

Keywords: Sustainable Tourism, System Dynamics, Vensim, Sensitivity Analysis, RESET Framework, Environmental Governance

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1 Introduction

1.1 Background

As a world - famous tourist spot, Juneau, Alaska, has drawn many visitors in recent years. In 2023, it hosted a record - breaking 1.6 million cruise passengers, with daily visitor numbers up to 20,000. The thriving tourism industry brought the local economy about 375 million dollars in revenue. However, this growth has also put great pressure on local social and environmental resources. The increase in tourists, environmental consumption, social pressure, and economic gains form a complex cycle of positive and negative feedback.

In this situation, the quality of life for residents and the sustainability of tourism resources face big challenges. According to system dynamics theory, balancing tourism development with the carrying capacity of social and environmental resources is key to ensuring long - term sustainable tourism in Juneau. Specifically, the local government needs to design and implement tourism policies that promote industry growth while minimizing negative impacts on society and the environment

1.2 Restatement of the Problem

In order to help the government seek a balance between tourism, the environment, and society, we need to solve the following questions:

- Develop a sustainable tourism model for Juneau,Alaska, explaining how factors interact through feedback mechanisms within the system, identifying key variables prioritized for optimization, and specifying critical constraints that define operational boundaries.
- Perform a sensitivity analysis to evaluate the model's robustness and determine the key factors driving system outcomes.
- Explain how to apply this model to other overtourism regions, adapting it to local needs.
- Explain how to adapt this model to less-visited regions or attractions, achieving a balance between tourism growth and sustainable resource management.
- Write a one-page memo for the Juneau Tourism Board, outlining your predictions, the effectiveness of various measures, and your recommendations on how to optimize outcomes.

1.3 Our Work

In section 1.2, we identified the key issues that need to be addressed in order to help Juneau, Alaska achieve a balance in sustainable tourism. To address these issues, our work mainly includes the following aspects:

- Developing a Sustainable Tourism System Dynamics Model: Using System Dynamics, we analyze Juneau's tourism trends and interactions between visitor numbers, resource consumption, social pressures, and revenue. The model aims to balance tourism growth with local social and environmental sustainability.
- Sensitivity Analysis and Policy Optimization: Identify key factors affecting Juneau's tourism sustainability and important variables via sensitivity analysis. Propose policy optimization strategies to minimize environmental and social impacts while visitor numbers grow.

- Model Promotion and Adaptability Analysis: Explore applying Juneau's sustainable tourism model to other over - tourism - affected regions and designing adaptive strategies based on regional characteristics, focusing on resource management and sustainable development for less - visited areas.
- Designing and Evaluating Policy Measures:Design policies like visitor limits and tax adjustments based on model results to address overcrowding, and evaluate their long - term effectiveness on Juneau's tourism industry.

2 Preparation of the Models

2.1 Assumptions

***Assumption* 1:** **Dominance of Key Variables**
Explanation: The dynamic changes in the tourism system are primarily driven by visitor density, infrastructure capacity, and ecological recovery rates, while secondary factors (e.g., extreme weather events, public health crises) are excluded.

***Assumption* 2:** **Fixed System Boundaries**
Explanation: The model assumes no external macro-level disruptions (e.g., global economic recessions, international energy price fluctuations) and focuses solely on internal tourism system dynamics (visitor growth, infrastructure stress, ecological responses).

***Assumption* 3:** **Immediate and Full Policy Implementation**
Explanation: All policy interventions (e.g., visitor caps, dynamic pricing) are assumed to be 100% effective immediately upon enactment, with no time delays or implementation losses.

***Assumption* 4:** **Exclusion of Ecological Threshold Effects**
Explanation: The model assumes no critical thresholds in ecosystem degradation (e.g., sudden glacier collapse or species extinction), considering only gradual linear/nonlinear changes. Simplifies complex ecological dynamics for mid-term (510 year) predictability, based on historical data trends.

Additional assumptions are made to simplify analysis for individual sections. These assumptions will be discussed at the appropriate locations.

2.2 Notations

Some important mathematical notations used in this paper are listed in Table 1.

3 The preliminary establishment of the sustainable tourism development model

3.1 Determination of System Boundaries

A system is an assemblage of interacting components organically connected through material or information flows, designed to perform specific functions, encompassing all elements that significantly influence the system's characteristics.

Table 1: Notations used in this paper

Symbol	Description
TV	Tourist Volume
TR	Tourism Revenue
RHI	Resident Happiness Index
TNB	Tourism Net Benefit
RCI	Road Congestion Index
EDI	Environmental Degradation Index
TEI	Tourism Employment Index
GPI	Gross Product Index
EGI	Environmental Governance Investment
$PEGI$	Proportion Environmental Governance Investment
PI	Promotional Investment
PPI	Proportion Promotional Investment

*There are some variables that are not listed here and will be discussed in detail in each section.

Synthesizing existing research findings, this paper posits that the sustainable tourism industry of Juneau, Alaska, constitutes a comprehensive system, primarily involving five fundamental elements: Resources, Environment, Society, Economics, and Tourism itself, herein abbreviated as the RESET model.

The framework of the entire RESET model is illustrated in Figure 2.

3.2 Explanation of variables in the system model.

Drawing on the research findings of previous scholars [3] and the specific situation of Juneau, this paper selected 2 stock variables, 5 flow variables, and multiple auxiliary variables to simulate the internal causal feedback mechanism of the sustainable development system of Juneau's tourism industry, and analyzed the impact of external policy factors on the system. In order to focus on the core research issues, this paper gave priority to selecting key indicators such as the number of tourists, tourism revenue, congestion, and glacier melting volume, while ignoring some relatively less important factors. At the same time, compared with the previous research, this model further introduced new variables such as residents' happiness level and the proportion of environmental governance investment. Table 2 shows the main variables in the system model.

3.3 The establishment of the flow chart for the sustainable tourism model.

Based on the main conclusions and theoretical foundations from domestic and international research findings, this paper employs the professional system dynamics software VENSIM to construct the causal diagram among the main variables in the sustainable tourism model of Juneau, further clarifying the interactions and feedback relationships between the variables.The causal diagram is shown in Figure 3.

Through the causal diagram of the main variables in the sustainable tourism model of Juneau, we further constructed a system flow chart based on stock variables, flow rate variables, and partial auxiliary variables. For example, the number of tourists in the stock variables corresponds to the

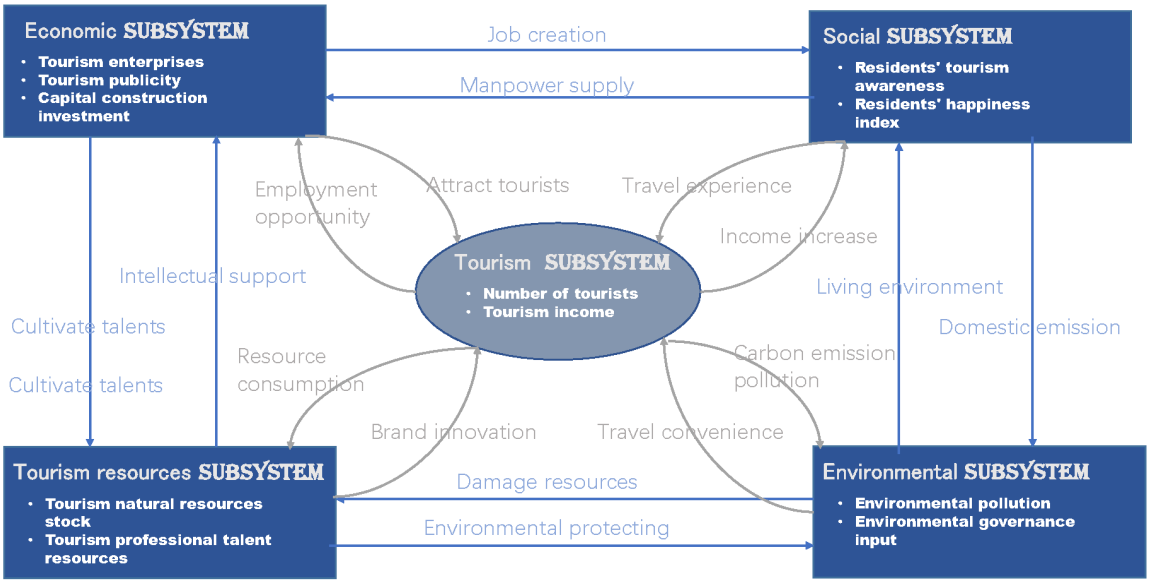


Figure 1: Model Overview

Table 2: Main variables of the System Dynamics Model for sustainable development of the tourism industry in Juneau.

Stock variables(L)	Flow rate variables(R)	Partial auxiliary variables(A)
TV	Tourist Number Increase	RHI, PEGI, Carbon Footprint
TNB	Tourist Number Decrease	RCI,Average Cost,TRS
-	TR	TEI,Tax Rate Increase,EDI
-	EGI	GDP,PPI,Glacier Melting Volume
-	EGI	GPI,Resident Tourism Awareness,Tempetature

*The abbreviations mentioned in the variables are reflected in Table 1.

increase and decrease in the number of tourists in the flow rate variables, as well as the tourism resource stock, the proportion of promotional investment, residents' tourism awareness, carbon footprint, and so on in the partial auxiliary variables. Through the analysis of the system flow chart, we can clearly recognize the inter - restraining and inter - influencing relationships between different variables in the model.The system flow chart we constructed is as shown in Figure 4.

3.4 Main Feedback Loop Analysis

The system flow chart for the sustainable development of Juneau's tourism industry is constructed based on the system causal diagram. Through the VENSIM software, we are able to obtain all the feedback loops in the system. Due to the numerous relationships between variables, this paper selects seven main logical chains that affect the number of tourists for analysis:

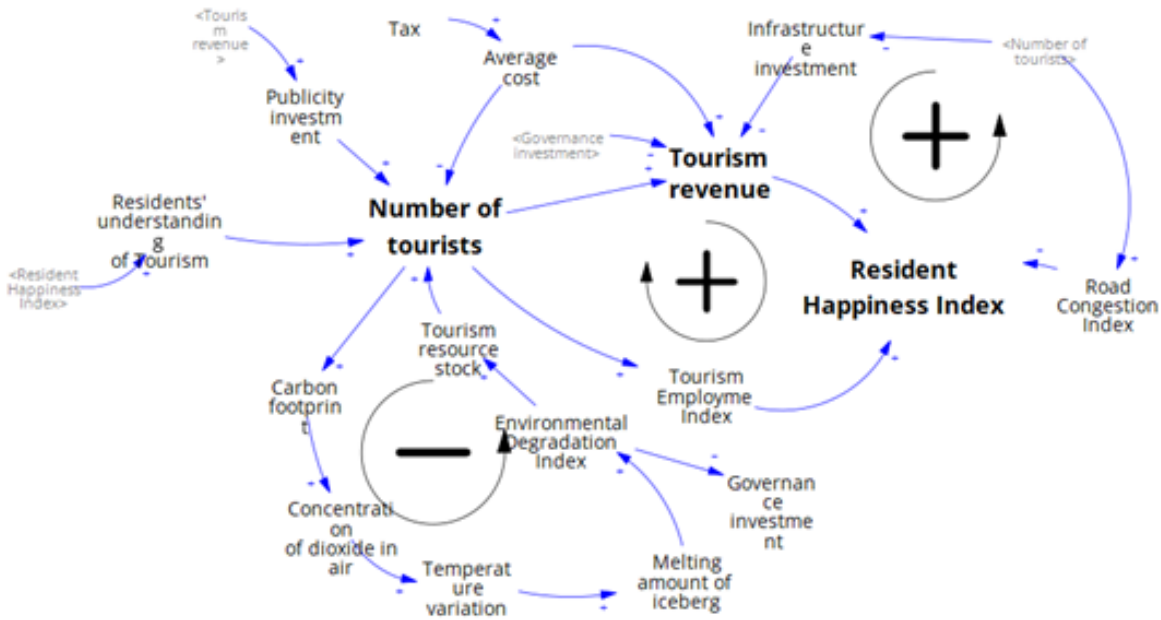


Figure 2: Casual Chart

- TV - Tourism Revenue (TR) - Tourism Employment Index (TEI) - Gross Product Index (GPI) - TV
- TV - Road Congestion Index (RCI) - Resident Happiness Index (RHI) - TV
- TV - Infrastructure Investment - TR - GPI - Happiness Index - TV
- TV - Carbon Footprint - Concentration Increment (ppm) - Temperature Change - Glacier Melting Volume - Environmental Degradation Index (EDI) - TV
- Tax Rate Increase - TV
- Tax Rate Increase - Average Cost - TR - TEI - RHI - TV
- Environmental Governance Investment (EGI) - Income - RHI - Resident Tourism Awareness - TV

The tourism industry in Juneau focuses on the number of tourists (TV), which is affected by feedback mechanisms. Positive feedback boosts TV via economic growth, infrastructure, and environmental governance: TV growth raises TR and employment, allowing government investment in infrastructure and environmental protection, enhancing service quality and residents' happiness, and drawing more tourists. Though long - term environmental investment may temporarily cut disposable income, ecological gains boost destination appeal and sustainable growth. Negative feedback, however, curbs unchecked expansion: TV spikes lead to congestion and resident discontent, environmental decline (e.g., glacier melt) dims tourism appeal, and tax hikes raise costs, hurting competitiveness. These factors create a dynamic balance where economic gains and infrastructure drive growth, while environmental limits and resident well - being set natural bounds. Policies should foster a green economy and community ties, with dynamic thresholds and flexible taxes to ensure TV rises steadily within ecological and social limits

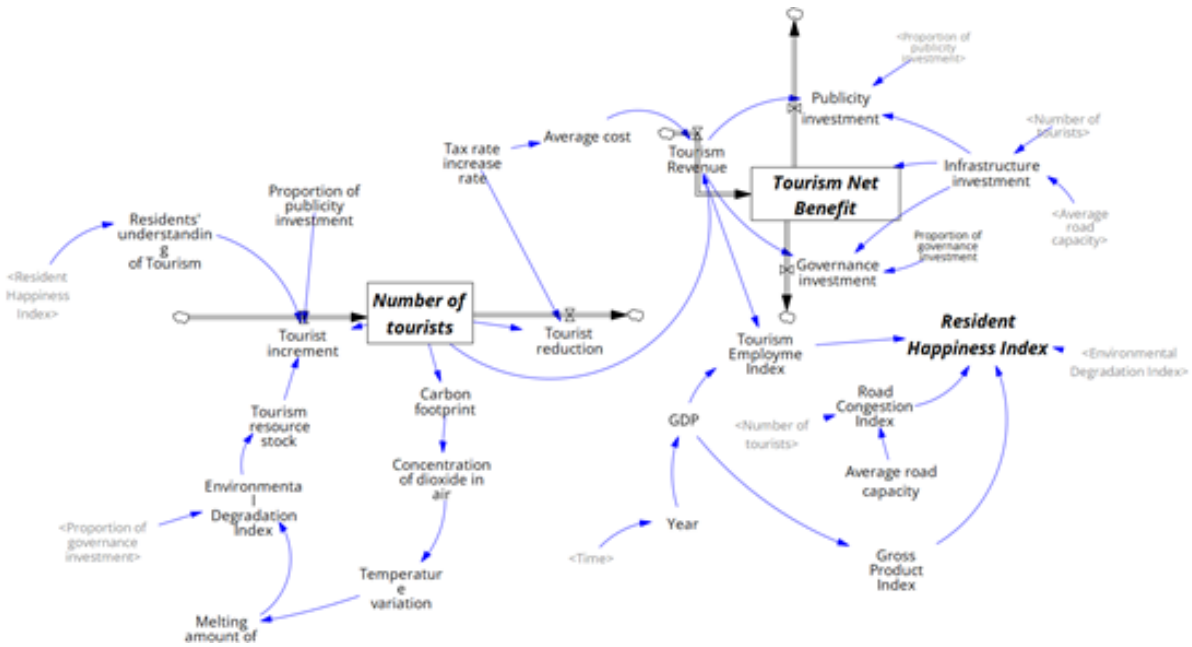


Figure 3: Flow Chart

4 Function Relationship Determination and Validation of the Sustainable Tourism Model

4.1 Variable Assignment and Function Relationship Determination

After constructing the system flowchart and analyzing the feedback loops, we proceed to determine the functional relationships between the major factors in the system through regression fitting and table functions.

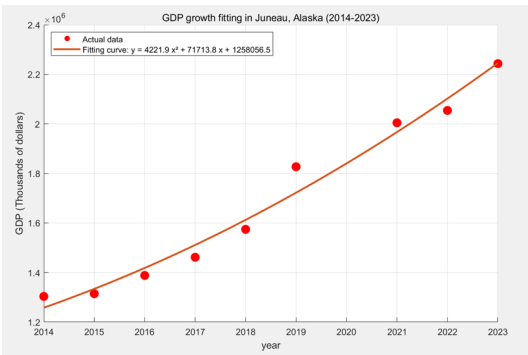
4.1.1 Function Fitting Based on Queryable Historical Data

We have obtained various official data provided by the City of Juneau, such as GDP, TNB, and others. To precisely fit the data, we used Python for curve fitting. Below is an example code that demonstrates how to perform regression analysis in Python to determine the relationship between tourism revenue and tourist numbers.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit

# Data and fitting function
x_data, y_data = np.array([1, 2, 3, 4, 5, 6])
np.array([2.1, 3.9, 6.2, 8.1, 9.8, 12.3])
def linear_func(x, a, b): return a * x + b

# Perform the fit
params, _ = curve_fit(linear_func, x_data, y_data)
```

```
# Visualization
plt.scatter(x_data, y_data, label='Data')
plt.plot(np.linspace(1, 6, 100)
linear_func(np.linspace(1, 6, 100), *params)
label='Fit', linestyle='--')
plt.legend(); plt.show()

# Output the results
print(f"y = {params[0]:.2f}x + {params[1]:.2f}")
```

Considering the profound impact of the COVID-19 pandemic on economic activities, this study has chosen not to analyze or discuss data from 2020. Next, we explore the relationship between year and tourism revenue. By using data from the City of Juneau's official website and FRED, and combining Python for data analysis and fitting, we obtained the relationship curve between the year and Juneau's GDP.



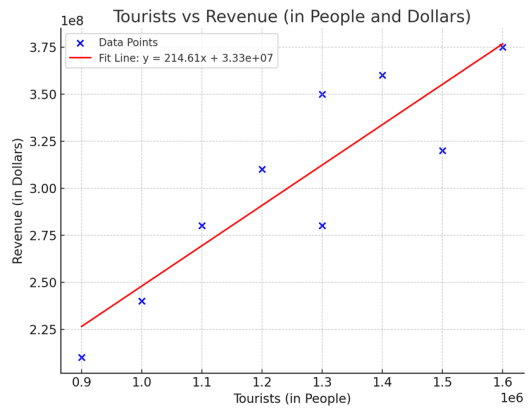


Figure 4: Relationship between Year and Juno City GDP and Relationship between Tourist Volumn and Tourism Revenue

Using the least squares method for quadratic fitting (figure4), the following fitting function was obtained:

$$f(x) = 4221.9x^2 - 71713.8x + 1258056.5$$

Where $f(x)$ represents Juneau's GDP in thousands of dollars, and x is the normalized year variable, defined as $x = \text{year} - 2014$.

Furthermore, we analyzed the relationship between tourist numbers and tourism revenue.

By using data from the Juneau government and economic reports, and Python for fitting (figure 4), we obtained the following linear relationship model:

$$f(x) = 214.61x + 3.33 \times 10^7$$

Where x represents the number of tourists (in persons), and $f(x)$ represents the corresponding tourism revenue (in dollars). This linear regression model indicates that for each additional tourist, tourism revenue increases by approximately 214.61 dollars.

4.1.2 Function Fitting Based on Literature and Assumptions

We begin by analyzing the residents' happiness index:

$$\begin{aligned}\text{RHI} = \text{GPI} \times a + \text{Social Health Index} \times b + \text{Social Welfare Index} \times c \\ + \text{Social Civilization Index} \times d + \text{Ecological Environment Index} \times e \quad (1)\end{aligned}$$

Where the weight coefficients a, b, c, d, and e represent the relative importance of GDP, social health, social welfare, social civilization, and ecological environment indices in residents' happiness.

Based on the U.S. national conditions and the analysis above, the preliminary weight settings are:

$$a = 30\%,\ b = 20\%,\ c = 25\%,\ d = 15\%,\ e = 10\%$$

In the context of sustainable tourism models, we modify the impact factors on residents' happiness index, introducing congestion index, environmental degradation index, tourism employment index, and GDP index. Since the indices mentioned are all percentages, we convert the addition operation into multiplication and assign indices to adapt to the model. Based on the data and literature analysis of Juneau, the final weight indices are:

$$\text{Congestion Index} = 0.95,\ \text{Environmental Degradation Index} = 1.1$$

$$\text{GDP Index} = 1,\ \text{Tourism Employment Index} = 0.5$$

Thus, the final formula for calculating residents' happiness is:

$$RHI = RCI^{0.95} \times EDI^{1.1} \times GDP^{1} \times TEI^{0.5}$$

The visual model is shown in Figure 5

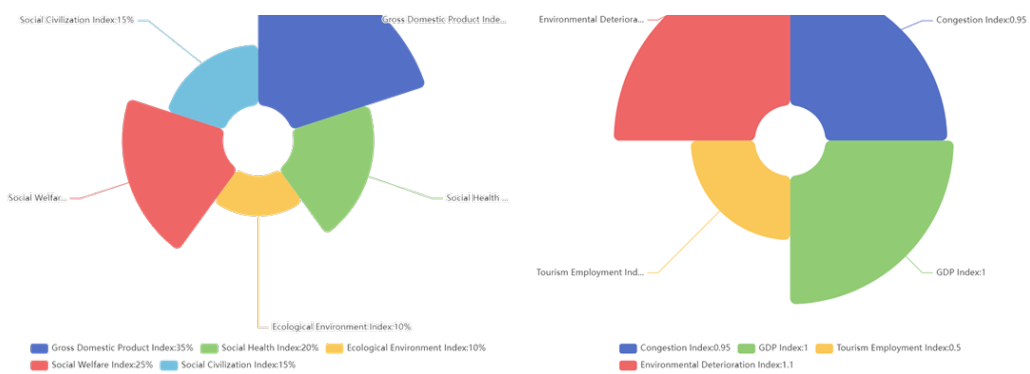


Figure 5: Initial and Modified Weight of the Resident Happiness Index

The impact of four indices on residents' happiness is considered:

Road Congestion Index (RCI): The Road Congestion Index is a mediating variable for the impact of tourist numbers on residents' happiness. While an increase in tourists brings more revenue, it

also affects the normal life of local residents. Here, we consider the traffic-related impacts due to excessive numbers.

According to the Federal Highway Administration (FHWA) publication on "Roadway Design Handbook" , the lane width typically ranges from 9 to 10 feet (approximately 2.7 to 3.0 meters), and the capacity of a single lane is about 1,500 vehicles per hour. For a major road with four lanes, the total capacity is about 6,000 vehicles per hour.

Assumptions:

- Local traffic constitutes one-third of total traffic.
- Tourist traffic increases by 500 vehicles for every additional 1,000 tourists.

Congestion Definition:

- Light Congestion: Traffic flow nears capacity, vehicle speed drops, but cars still move slowly. The flow typically reaches 80%-90% of road capacity.
- Heavy Congestion: Traffic exceeds capacity, vehicle speed becomes very low, or even stalls, causing severe vehicle queuing. The flow exceeds 90%-100% of capacity.

For the tourist flow in the peak season, assuming the total annual tourists is N, the formula for peak season traffic is:

$$\frac{(N \times 0.9)}{210 \times 0.5 + 2000} / 6000$$

For the impact on happiness, the RCI based on congestion level P is divided into sections (with 100% as the baseline):

$$RC_1 = \begin{cases} 11 & \text{if } P < 68\% \\ 0.9880 & \text{if } 68\% < P < 80\% \\ 1 - (P - 0.8) & \text{if } 80\% < P < 90\% \\ 1 - (P - 0.8) \times 1.02 & \text{if } P > 90\% \end{cases}$$

Gross Domestic Product Index (GPI): The GDP index is an important indicator of the national economic level. The formula for calculating it is:

$$GPI = \frac{\text{GDP}}{\text{delay1(GDP,1)}} \times 100\%$$

Where the delay function in Vensim is used to define the previous year's GDP.

Environmental Degradation Index: Tourism-related traffic is a major source of carbon emissions in the tourism industry, accounting for over 80% of its total carbon emissions [**?**]. The estimated increase in carbon emissions due to a rise in tourist numbers is modeled as:

$$\text{Carbon footprint} = TV \times 0.2651325$$

To convert the carbon footprint into atmospheric CO concentration increment (ppm), we use:

$$\Delta ppm = \frac{\Delta E}{M_{\text{atmosphere}}} \times 10^6$$

where ΔE is the additional carbon footprint, and $M_{\text{atmosphere}}$ is the total mass of the atmosphere (5.15 Œ $10^{15} tons$).

Carbon Footprint and its Impact on Regional Temperature Change

The increase in CO concentration leads to radiative forcing (RF), which is related to the CO concentration as described by the following equation:

$$F(t) = \alpha \ln \left(\frac{C(t)}{C_0} \right) \tag{2}$$

where $F(t)$ is the radiative forcing at time t, α is a constant taken as 5.35 W/mš, $C(t)$ is the CO concentration at time t, and C_0 is the initial CO concentration.

The relationship between radiative forcing and regional temperature change is expressed by the following equation:

$$\Delta T(t) = \beta F(t) \tag{3}$$

where $\Delta T(t)$ is the temperature change at time t, and β is the climate sensitivity parameter, typically taken as 0.8 K/(W/mš).

Combining the above formulas, we get the comprehensive model for the temperature change due to carbon footprint:

$$\Delta T(t) = \alpha\beta \ln \left(1 + \frac{\Delta C}{C_0} \right) \tag{4}$$

Juno, located in Alaska, is a relatively remote area, thus it is assumed that the initial CO concentration in Juno is close to the global background concentration, approximately 280 ppm.

Temperature Change and its Impact on Glacier Melting

The observation of glacier tongue melting and accumulation involves inserting several rods into the tongue surface, and measuring the change in elevation of the ice surface. When melting occurs, the ice surface decreases, and when accumulation happens, the ice surface increases. The average melting or accumulation amount is obtained by averaging the values from all the rods.

Through observation and fitting, it is known that:

$$T = 1.72 + 0.88t \tag{5}$$

where t represents the 600 hPa temperature in degrees Celsius, and T is the temperature at the glacier tongue's end in degrees Celsius.

The melting amount Y is related to T by the following equation:

$$Y = 0.4 + 4.8T \tag{6}$$

Since the model includes natural annual melting, the average natural melting per year is 8.656 mm. Thus, after subtracting this amount, the result can be used for further comparison.

The environmental degradation index (EDI) is given by:

$$EDI = Y \times 10^6 \times 100\% \tag{7}$$

Using the temperature change model, we can estimate the relationship between the environmental degradation index and tourist numbers.

Tourism Employment Index

The tourism employment index reflects the number of tourism-related jobs in Juno, which acts as an intermediary variable between tourism revenue and residents' well-being. Using the proportion of tourism revenue to GDP, we can derive the tourism employment situation. We also use a delay function to calculate the index, which compares the ratio with the previous year. The formula for the tourism employment index is as follows:

$$TEL = \frac{TR}{GDP} \Big/ \text{delay1}\left(\frac{TR}{GDP}, 1\right) \times 100\% \tag{8}$$

Impact of Tourist Numbers on Infrastructure Investment (Water and Waste)

Water Increase:

$$\Delta W_d = N_V \times W_p \tag{9}$$

where ΔW_d is the increase in daily water demand (in liters per day), N_V is the number of tourists, and W_p is the per capita daily water usage (in liters per person per day).

Waste Increase:

$$\Delta G_d = N_V \times G_P \tag{10}$$

where ΔG_d is the increase in daily waste production (in kg per day), N_V is the number of tourists, and G_P is the per capita daily waste production (in kg per person per day).

Total Infrastructure Investment: The total infrastructure investment cost is calculated as:

$$l_t = (C_W \times \Delta W_d) + (C_g \times \Delta G_d) \tag{11}$$

where l_t is the total cost (in USD), C_W is the water supply and treatment cost per cubic meter (in USD per cubic meter), and C_g is the waste treatment cost per ton (in USD per ton).

Additionally, the infrastructure investment formula should include road expansion costs due to

the impact of congestion:

$$l_t = (C_W \times \Delta W_d) + (C_g \times \Delta G_d) + (W - W_0) \times 1.5 \times 10^4 \tag{12}$$

where W is the average road capacity, and W_0 is the initial average road capacity (6000 vehicles/hour as assumed for Juno's four-lane road).

Parameter Reference Values:

- C_W: Water treatment cost per cubic meter is approximately 0.5-1.5 USD (depending on the region).
- C_g: Waste treatment cost per ton is approximately 50-100 USD.
- W_p: Per capita daily water usage is 100-300 liters per person per day.
- G_P: Per capita daily waste production is 1-2 kg per person per day.
- W_0: The initial average road capacity is assumed to be 6000 vehicles per hour.

Tourism Revenue, Promotion Investment, and Tourist Numbers

To quantify the impact of promotional investment on tourism revenue, we construct a simple model to describe the relationship between promotional investment, tourist numbers, and revenue. The model assumptions are as follows:

- The relationship between promotional investment and tourist numbers follows a logarithmic growth. That is, early increases in promotional investment significantly increase tourist numbers, but the marginal benefit decreases as investment increases.
- The relationship between tourist numbers and tourism revenue is linear, assuming each tourist has a constant average spending amount.

Tourist Numbers: The relationship between tourist numbers N_v and promotional investment I is given by:

$$N_v = N_0 \times \ln(I + e) \tag{13}$$

where N_v is the number of tourists, N_0 is the baseline number of tourists (with no promotional investment), and I is the promotional investment.

Total Revenue: The relationship between total revenue R and tourist numbers N_v is given by:

$$R = Sp \times N_v \tag{14}$$

where Sp is the average spending per tourist.

By combining the two equations, we can express total revenue as:

$$R = Sp \times N_0 \times \ln(I + e) \tag{15}$$

Simplifying, we get:

$$R = k \times \ln(I + e) \tag{16}$$

where $k = Sp \times N_0$ is a combined coefficient reflecting the product of the baseline tourist number and per capita spending.

Parameter Explanation:

- N_0: The baseline number of tourists, which can be estimated from historical data.
- Sp: The average spending per tourist, including accommodation, dining, entertainment, etc.
- I: The promotional investment ratio, which includes advertising, marketing activities, etc.
- k: A coefficient reflecting the combined influence of baseline tourist numbers and per capita spending.

The constructed models help analyze the interactions between key factors such as tourism revenue, promotional investment, tourist numbers, and the ecological environment, providing data support for the sustainable development of tourism in Juno, and offering theoretical foundations for future policy optimization and environmental protection.

4.2 Model Verification

To verify the accuracy and reliability of the developed model, three verification methods were employed in this study:

- **Visual Inspection:** The model's boundaries, variable definitions, types, causal loops, and equations were examined to ensure structural rationality
- **Validity Test:** Simulation results (2014 - 2023) were compared with Juneau's actual tourism data. Key variable relative errors assessed model predictability.
- **Stability Test:** Step size testing confirmed model stability. Consistent trends across different step sizes indicated no pathological results.

4.2.1 Visual Inspection

Using the model's Check Model and UnitsCheck, we examined whether the model structure was reasonable, whether the statements were correct, and whether the units were consistent. Additionally, by further analyzing the system's causal relationships and SD flow diagrams, we checked if the models system structure and driving mechanisms were rational. After inputting the 2014 tourism population of 900,000 in Juneau, the model ran normally without any errors, confirming the success of the visual inspection.

4.2.2 Validity Test

Based on the developed system dynamics model for sustainable tourism development in Juneau, Vensim PLE software was used to simulate the historical data of key tourism indicators in Juneau from 2014 to 2023.

Year	Predicted Tourists	Actual Tourists	Relative Error
2014	900,000	900,000	0.00%
2015	1,012,360	1,000,000	1.24%
2016	1,106,670	1,100,000	0.61%
2017	1,197,510	1,200,000	0.21%
2018	1,291,580	1,300,000	0.65%
2019	1,391,910	1,400,000	0.58%
2020	1,485,860	100,000	1385.86%
2021	1,532,190	1,300,000	17.86%
2022	1,533,600	1,500,000	2.24%
2023	1,515,380	1,600,000	5.29%

Table 3: Tourism Numbers for Juneau (2014-2023) and Predictions

Year	Predicted Revenue (USD)	Actual Revenue (USD)	Relative Error
2014	226,149,000	210,000,000	7.69%
2015	250,262,000	240,000,000	4.28%
2016	270,503,000	280,000,000	3.39%
2017	289,999,000	310,000,000	6.45%
2018	310,187,000	350,000,000	11.38%
2019	331,719,000	360,000,000	7.86%
2020	351,880,000	100,000,000	251.88%
2021	361,823,000	280,000,000	29.22%
2022	362,125,000	320,000,000	13.16%
2023	358,216,000	375,000,000	4.48%

Table 4: Tourism Revenue for Juneau (2014-2023) and Predictions

Comparing the simulation results with historical data in Table 3 and Table 4, the model showed a high degree of fit and effectively reflected the operational status of Juneau's tourism industry. This verified the validity and reliability of the system dynamics model established for the region.

4.2.3 Stability Test

By changing the time step in the model setting from 1 year to 2 years, the model stability was tested. Using the tourist volume and tourism revenue predictions from 2014 to 2023, we obtained two fitted curves as shown in Figure 6:

According to the principles of stability testing, we observed that the general trends of the model predictions remained consistent under different initial conditions with time steps of 1 year and 2 years. Furthermore, the relative differences between the predictions were within 10%, which indicates that the model passes the stability check.

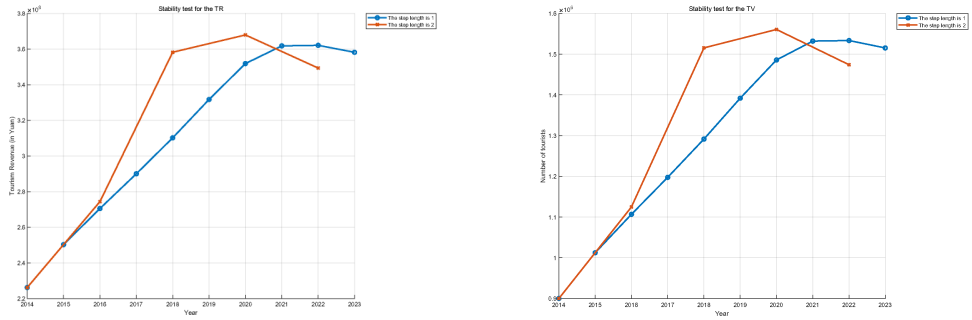


Figure 6: Stability Test for the TR and TV

5 Sustainable Tourism Model Sensitivity Analysis and Variable Optimization

5.1 Sensitivity Analysis

To more intuitively represent the impact of parameters on the total water demand, a sensitivity analysis of the model was conducted. The method for sensitivity analysis is to calculate the model's sensitivity, given by the formula:

$$S = \frac{\Delta X / X_0}{\Delta K / K_0}$$

Where: $\Delta X = |X - X_0|$, $\Delta K = |K - K_0|$, S is the parameter sensitivity; ΔX is the absolute change in the state variable caused by the parameter change; ΔK is the absolute value of the parameter change; X is the state variable after the change; X_0 is the initial value of the state variable; K is the parameter change; K_0 is the initial value of the parameter.

Table 5: Sensitivity Analysis Results

Parameters	+10%	-10%	Value Range	Sensitivity
TRI	0.093077	0.117242	0.05-0.2	High
PEGI	0.009345	0.011421	0.009-0.05	Medium
PPI	0.003494	0.008527	<0.009	Low
Glacier Melting Index	0.000926	0.001132	<0.009	Low
Average Road Capacity	0.022501	-	0.009-0.05	Medium

Note: Since reducing the average road capacity is unrealistic, it is not considered.

The sensitivity analysis in Table 5, it shows that the tax rate parameter has a significant impact on the model's results, followed by environmental investment share and average road capacity.

The order of sensitivity is: **Tax Rate > Average Road Capacity > Environmental Investment Share > Publicity Investment Share > Glacier Melting Index**. From the sensitivity analysis of different parameters, it can be concluded that when modifying policies, more attention should be given to variables with high sensitivity, while also considering practical feasibility.

5.2 Variable Optimization

After sensitivity analysis, we chose variables with medium and high sensitivity for optimization: tax rate increment, environmental investment share, and road capacity. Since residents' well-being combines economic (tourism revenue), environmental (environmental deterioration index), and social (congestion index) factors, it reflects the overall policy impact on the local RESTET model. Thus, it's used as a comprehensive evaluation index, with the variable control method as the scientific approach.

In road capacity optimization, we ran simulations by adjusting road capacity (keeping other parameters constant). Results in Figure 9 showed residents' well-being follows a quadratic curve with road capacity changes, first rising then falling. When road capacity is low, the congestion index rises, cutting traffic efficiency and harming residents' well-being. But if road capacity is too high, though congestion eases, increased government spending and lower resource efficiency also hurt residents' well-being. So, both too low and too high road capacities harm residents' well-being.

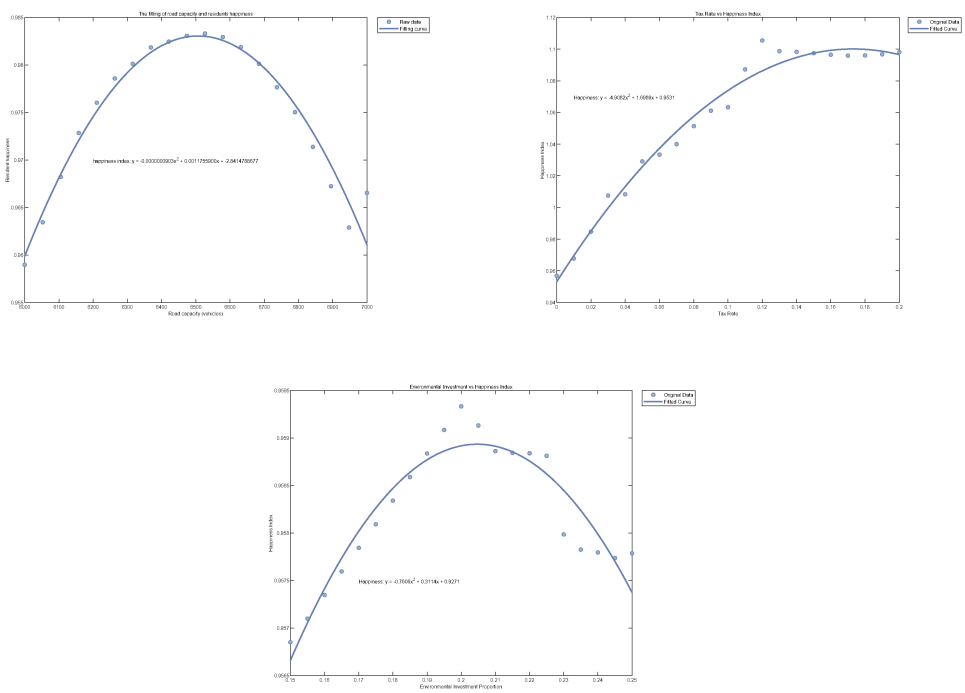


Figure 7: The Relationship between Road Capacity, Tax Rate, Environmental Investment, and Residents' Well-being

Simulation analysis found the optimal road capacity range is 6400 - 6600 vehicles/hour. We suggest Juneau set road capacity at 6500 vehicles/hour to boost residents' well-being and support sustainable tourism goals

In tax rate optimization, simulations adjusted the tax rate while keeping other parameters constant. Results in Figure 7 showed the relationship between residents' well-being and the tax rate first increases and then stabilizes, following a quadratic curve. When the tax rate is low, insufficient government revenue limits public services and social welfare, affecting residents' well-being. As the tax rate rises, public services and social welfare improve, significantly boosting residents' well-being. However, when the tax rate reaches a certain level, further increases lead to diminishing marginal returns on well-being and may suppress economic vitality and residents' consumption due to excessive tax burdens.

Simulation analysis found the optimal tax rate range for residents' well-being to be 0.16 - 0.18. It is recommended to adjust the tax rate to 0.17 to balance government revenue and social development needs, promoting sustainable development goals.

In optimizing environmental governance investment, simulations adjusted the investment share while keeping other parameters constant. Results in Figure 9 showed the relationship between residents' well-being and environmental investment follows a diminishing marginal return pattern. Low investment leads to poor ecological quality, negatively affecting well-being. Increasing investment significantly improves well-being, but further increases yield diminishing returns. If the investment share is too high, other social resources may be insufficient, hindering well-being improvement.

Simulation analysis found the optimal investment share range for residents' well-being to be 0.19 - 0.21, with the best ratio being 0.20. It is recommended to optimize the environmental governance investment share to 0.20 to balance ecological protection with social and economic development, contributing to sustainable development goals.

5.3 Verification of the Effectiveness of Variable Optimization

The optimization of key variables such as the tax rate, environmental investment share, and road capacity aims to balance sustainable tourism development and residents' well-being in Juneau. This section presents the effects of these optimizations in achieving the expected goals. The figure 8 clearly reflect the simulation results.

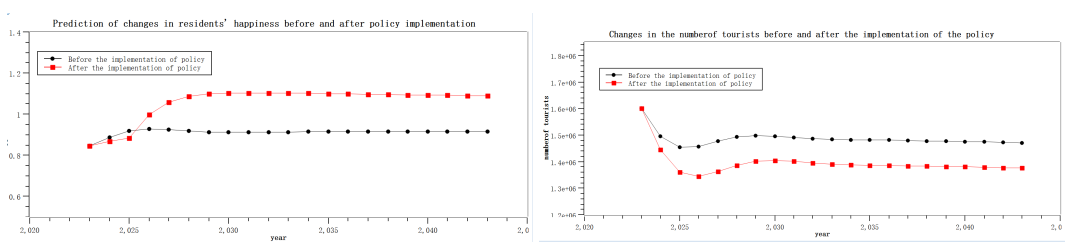


Figure 8: Changes in Residents' Well-being and Tourist Numbers

5.3.1 Changes in Residents' Well-being

From the first figure, it can be seen that the optimization of key variables has had a positive impact on residents' well-being. The red line (after policy implementation) shows a significant upward trend, while the black line (before policy implementation) remains steady. The improvement in well-being reflects the combined effects of tax policy adjustments, enhanced environmental governance, and infrastructure improvements, indicating that our policies have achieved the goal of enhancing residents' quality of life.

5.3.2 Changes in Tourist Numbers

The second figure shows the changes in tourist numbers before and after policy implementation. It can be observed that after the policies were implemented (red line), the number of tourists decreased and stabilized, compared to the uncontrolled growth (black line) before the policies. This control ensures that the number of tourists remains at a more sustainable level, preventing the tourism industry from exceeding local resource capacity and damaging residents' quality of life.

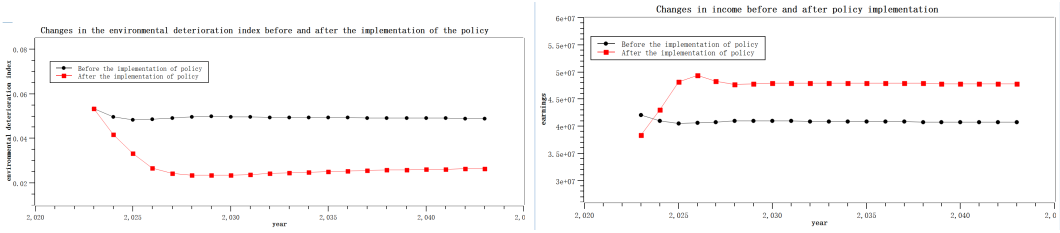


Figure 9: Changes in Environmental Degradation Index and Tourism Revenue

5.3.3 Changes in Environmental Degradation Index

The figure 9 shows the trend of the environmental degradation index. After policy implementation (red line), the environmental degradation index significantly decreased, indicating that the increase in environmental governance investment has led to better protection of the local ecological environment. Before the policy implementation (black line), the index remained high, further highlighting the importance of policy intervention in environmental protection.

5.3.4 Changes in Tourism Revenue

Finally, the fourth figure illustrates the impact of policy adjustments on tourism revenue. It can be seen that after policy implementation (red line), tourism revenue steadily increased. This growth is mainly due to the more sustainable and balanced tourism model, which, despite a decrease in the number of tourists, still results in higher-quality tourism services and increased economic benefits, while avoiding the overuse of regional resources.

In conclusion, through the optimization of key variables, we effectively enhanced residents' well-being, controlled tourist numbers, reduced environmental degradation, and achieved growth in tourism revenue. These results indicate that Juneau's tourism industry's sustainable development goals have been well supported, achieving a balance between economic and ecological benefits.

6 Sustainable Tourism Model Promotion

6.1 Destination Characteristics Analysis

When promoting the sustainable tourism model in different destinations, it is essential to conduct a comprehensive analysis of the unique characteristics of each destination. Factors such as natural resources, economy, culture, and infrastructure all have a profound impact on the sustainability of tourism.

6.1.1 Natural Resources and Environmental Characteristics:

Describe the natural resources of the destination, such as attractions, ecological environments, etc. Analyze the vulnerability of these resources and their impact on sustainable tourism, such as ecological carrying capacity and carbon emissions.

6.1.2 Economic and Social Characteristics:

Economic structure, income sources, and population characteristics of the destination. The proportion of tourism in the local economy and whether it depends on a single market (e.g., tourists from a specific country or seasonal tourists).

6.1.3 Cultural and Social Acceptance:

Cultural characteristics of the destination and the friendliness toward foreign visitors. The local community's acceptance of the sustainable tourism concept, and whether there is social recognition and support.

6.1.4 Infrastructure and Tourism Services:

The state of transportation, accommodation, and tourism services in the destination. How existing environmental protection measures and public facilities support sustainable development.

6.2 Model Adjustment Strategy

In the original model, many variables like congestion index, resident happiness, publicity investment ratio, and tourism employment are widely applicable and can be core variables for different destinations. Model adjustments should mainly focus on adapting to each destination's unique features, especially how environmental impact factors match the target area's conditions. For example, coastal destinations can introduce a seawater pollution index (P) to quantify impacts, while mountainous ones can introduce a geological disaster risk index (G) for accurate risk assessment. To enhance the model's practicality, surveys and data analysis will determine specific functions and formulas to assess these new variables' impact on the environmental index and other key variables. Combining system flow charts will help verify new functions and analyze their potential impact on resident happiness. Through these adjustments and validations, the model will more accurately reflect different destinations' realities, providing scientific support and decision - making assistance for sustainable development in various tourism areas.

6.3 Model Adaptation Example

6.3.1 Data Collection and Parameter Determination

In adapting the model for mountainous tourist areas, relevant data about the target region should first be collected, including tourist numbers, local tourism revenue, frequency and scale of geological disasters, regional population, infrastructure conditions, etc. This data will help determine the initial values and adjustment coefficients for each variable, providing foundational support for subsequent model validation and adjustments.

6.3.2 Model Adjustments

Based on the characteristics of mountainous tourist areas, the model needs to be adjusted as follows:

- Introduce Geological Disaster Risk Index (G): Due to the complex terrain of mountainous areas, geological disaster risk is one of the main challenges to tourism development. A Geological Disaster Risk Index (G) is introduced in the model to quantify the correlation between tourist numbers and geological disaster occurrences in the region. The expression is:

$$G = \gamma \cdot x + \delta \cdot h_x$$

where x represents tourist numbers, h_x represents the distribution of tourists in different hazardous areas, and γ and δ represent the coefficients for the impact of tourist numbers and tourist density in hazardous areas on geological disaster risk.

- Introduce Carbon Emissions (C): The construction and operation of transportation, accommodation, and other facilities in mountainous areas typically produce some carbon emissions. Therefore, the impact of tourist numbers and traffic on carbon emissions should be considered. The carbon emissions are related to tourist numbers and traffic density and can be described using the following expression:

$$C = \theta \cdot x + \zeta \cdot t$$

where x represents tourist numbers, t represents traffic density, and θ and ζ are coefficients representing the impacts of tourist numbers and traffic density on carbon emissions.

6.3.3 Model Solution and Result Analysis

After the model adjustments, we can derive the system flow chart for mountainous tourist areas along with the key variables. Then, using Vensim PLE software, the adapted model can be simulated, validated using historical data, and further subjected to error analysis and sensitivity analysis. Scenario simulations can then be used to propose corresponding policies, such as setting a tourist flow limit or implementing tourist dispersal based on the geological disaster risk index, reducing the concentration of tourists in high-risk areas, and strengthening infrastructure development, particularly for transportation, accommodation, and environmental protection, to enhance the carrying capacity and sustainability of the tourism industry.

6.4 Impact of Destination Selection on the Priority of Key Measures

The natural, economic, and social characteristics of a destination directly determine the priority of core measures in the sustainable tourism model. For resource - vulnerability - dominated areas (e.g., coastal or glacier tourist areas), focus on ecological carrying capacity management and pollution control, such as setting visitor limits, time - based visits, and introducing seawater pollution indices (PP) and dynamic carbon emission monitoring. For model adjustments, embed an "environmental degradation threshold" in the system dynamics, where the high - risk area closure policy is automatically triggered when pollution exceeds the threshold. Vensim simulations can optimize the flow limit based on the impact of different tourist distributions on pollution.

For economically dependent areas (e.g., small towns relying on a single tourism market), emphasize market diversification and community benefit - sharing, such as developing off - season tourism products, expanding cultural experience projects, and redistributing tourism revenue to residents. The model should add a "market diversity index" to quantify off - season tourist contributions and introduce a "resident participation" variable to simulate the effect of dividend - sharing policies on resident support

For risk-sensitive areas (e.g., mountainous or geological disaster-prone areas), safety management and low-carbon transportation upgrades are key. The model should dynamically link tourist distribution and risks via the geological disaster risk index $G = \gamma \cdot x + \delta \cdot h_x$ (where x is the number of tourists, and h_x is the tourist density in dangerous areas, with γ and δ fitted through historical data), while promoting alternative transportation options like cable cars and ecological trails to reduce self-driving. The model should also link traffic modes (e.g., proportion of self-driving) with carbon emissions (C) and optimize low-carbon transportation subsidy policies to reduce environmental pressure.

6.5 Using the Model to Promote Low-Visitor Flow Attractions and Achieve Regional Balance

System dynamics models optimize tourist distribution via key measures: differentiated publicity, infrastructure linkage, off-season incentives, dynamic pricing, community collaboration, revenue redistribution, carrying capacity balance, and risk early warning. For publicity, adjust weights to favor less - visited sites (e.g., 20% budget to eco - villages) and add a "less - visited publicity weight" variable. Vensim simulates tourist growth under varying weights to find the optimal ratio. Shuttle buses between popular and less - visited sites enhance accessibility; a "transport accessibility index" quantifies their impact (e.g., 50% more shuttles could boost less - visited site visitors by 25%).

For off-season incentives, tiered ticket pricing (e.g., 20% discount for off-season tickets) and "popular + less-visited" combined ticket packages should be designed. A "off-season discount-tourist volume" function (e.g., $N_v = N_0 \cdot \ln(1 + \beta \cdot D)$, where D is the discount rate and β is fitted using historical data) should be built in the model. The simulation shows that the discount strategy could increase visitor volume by 15%-20%. For community collaboration, the "community promotion index" should be introduced to quantify the impact of residents' participation in homestay or cultural projects on tourist growth (e.g., a 10% increase in resident participation could lead to an 8% increase in tourist volume). Through fiscal transfer payments, 30% of the tax revenue from popular attractions should be reinvested into infrastructure in less-visited regions. The model should use the "fiscal transfer ratio" variable to assess its balancing effect (e.g., a 5% increase in the transfer ratio could lead to a 3% annual increase in tourist volume).

Carrying capacity balance should set ecological thresholds (e.g., maximum visitor count of 500 per day) in the less-visited attractions model. Exceeding the threshold should trigger automatic flow control. Real-time data platforms should dynamically monitor tourist distribution and carbon emissions, with quarterly updates to model parameters to avoid overdevelopment. For example, when the visitor volume of a less-visited attraction approaches the threshold, its publicity weight should be automatically reduced to ensure sustainable development.

7 Memorandum to the Juneau Tourism Board

7.1 Current Situation and Issues

Tourism in Juneau has grown rapidly in recent years, but significant challenges have emerged. During peak season, tourists make up 90% of the traffic, causing frequent congestion, with traffic flow on major roads often reaching or exceeding capacity, and the congestion index staying above 80% for long periods. This directly affects residents' quality of life and visitor experience. Meanwhile, tourism activities have increased carbon emissions, with the per capita carbon footprint growing by about 0.265 tons annually, speeding up glacial melting. Model predictions show that without intervention, glacial melting could increase by 12% in the next decade, threatening the local ecosystem's stability. Despite rising tourism revenue, residents' happiness has been negatively affected by traffic congestion and environmental degradation. Some communities oppose tourism expansion. Current policies have flaws, such as the tax rate (0.15) failing to support public service investments, and the insufficient share of funds for environmental governance (10%), making it hard to balance ecological protection and economic development.

7.2 Model Prediction Results

Based on the simulation analysis of the system dynamics model, if current policies remain unchanged, by 2030, the congestion index will exceed 90%, residents' happiness may decline to 85% of the baseline value, and glacial melting will increase by 1.5% annually, significantly threatening ecosystem stability. However, if optimization measures are implemented, such as raising the tax rate to 0.17, residents' happiness could improve by about 15%, while government revenue would increase by 8%. Expanding road capacity to 6,500 vehicles per hour could control the congestion index below 75%, boosting resident satisfaction by 10%. Increasing environmental governance funding to 20% would slow the carbon emission growth by 30%, and the annual increase in glacial melting could be reduced to 1.0%. These results demonstrate that policy adjustments can significantly improve the sustainability of tourism.

7.3 Evaluation of Existing Measures

Current policies have revealed several deficiencies in their implementation. The low tax rate (0.15) results in insufficient funding for public services (such as waste management and drinking water supply), leading to infrastructure operating costs exceeding the budget by 30%. Road capacity (6,000 vehicles per hour) has not reached the optimal range (6,400-6,600 vehicles per hour), resulting in severe congestion during the peak season, exacerbating residents' complaints. Environmental protection measures lack targeted strategies, with funds being dispersed, leading to an annual decline of 2% in the ecological index. Additionally, current promotional strategies have not effectively enhanced residents' recognition of sustainable tourism, with community support at only 62%, reflecting a disconnect between policies and social needs.

7.4 Optimization Recommendations

The recommended priority is fiscal and tax reforms, gradually increasing the tourism - related tax rate to 0.17. This revenue should fund public services, community welfare, and a "Sustainable Tourism Fund" for environmental governance, focusing on carbon capture and glacier monitoring. Infrastructure improvements include expanding main roads to four lanes (6,500 vehicles per hour) and implementing smart traffic management. Invest in eco - friendly facilities (e.g., EV chargers and solar - powered accommodations) to reduce tourism's carbon footprint.

To encourage off - season visits, lower peak - season tourist proportion from 90% to 75% via discounts and a "Green Juneau" campaign.

Establish a real - time data platform to monitor tourists, emissions, and glacier melting, and publish an annual report.

Create a task force for ongoing policy assessments to ensure flexibility and adaptability.

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AI Use Report

In the process of writing this paper, the team utilized AI to refine and improve expressions that were deemed awkward or non-native. All AI-generated language refinements were thoroughly reviewed and approved by the team before being applied to the respective sections of the paper. Additionally, any AI-generated text was produced based on well-defined prompts and adequate content, ensuring that the team did not engage in the misuse of AI. We emphasize that the AI was used solely for language improvement purposes, and all content was carefully scrutinized by the team to maintain the integrity and quality of the paper.